

Mini Report - Exposed Rebar Inspection with YOLOv8

Detecting visible exposed reinforcement in concrete-inspection images to flag cases for human condition assessment

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1. Problem, use case and human role

Problem. Concrete-inspection photographs can contain visible exposed reinforcement, but manually reviewing many images is slow and may overlook cases that need attention.

Use case. A computer-vision model identifies visible exposed_rebar in inspection images and flags the image for review by a qualified engineer. The output is a bounding box and a segmentation mask with a confidence score.

Human role. The engineer reviews every flagged image and decides whether further condition assessment, testing or repair design is needed. A detection is an inspection flag, never a diagnosis.

Error cost. A miss (false negative) is treated as roughly three times more costly than a false alarm: unflagged exposed reinforcement may never be inspected, whereas a false alarm only adds one human review.

2. Target class and labelling rules

Item	Positive definition	Boundary cases	Box / mask rule
exposed_rebar	A reinforcing steel bar visibly exposed where the concrete cover is absent or broken.	Do not label cracks, including cracks parallel to reinforcement or with rust staining, unless a distinct visible steel bar can be seen.	One tight box around each separate continuous visible rebar segment. Include only visible steel; exclude surrounding concrete, voids, rust stains and shadows. Minimum target size 20 × 20 px; do not label smaller or unclear objects.

3. Method and training configuration

Model: YOLOv8n-seg (Ultralytics) instance segmentation, trained on the Roboflow dataset rebar-exposure-qm020 v4 (1,547 training / 168 validation images). Training: 30 epochs, image size 640 × 640, batch size 8, random seed 42, NVIDIA Tesla T4 GPU, approximately 20 minutes. The selected checkpoint best.pt, the training curves (results.png), the epoch-level metrics (results.csv) and the Colab notebook are stored in the project repository, so the run can be repeated end to end.

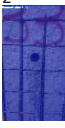
4. Validation results and interpretation

Metric	Bounding boxes	Segmentation masks	Reading
Precision	0.805	0.808	About 1 in 5 flags is incorrect.
Recall	0.703	0.717	About 28 % of annotated instances are missed — the main risk.
mAP@50	0.772	0.762	Reliable at a loose overlap threshold.
mAP@50–95	0.529	0.470	Masks are coarse at strict overlap thresholds.

Validation set: 168 images, 229 annotated instances. Mask metrics are the primary result, because the task requires spatial delineation and not only localisation. The gap between mAP@50 and mAP@50–95 is the quantitative counterpart of what the qualitative review below shows: the model finds the right region but draws it too generously.

5. Error analysis on nine new inspection images

The trained checkpoint was applied to nine photographs outside the training workflow. Each prediction was reviewed visually against the labelling rules of Section 2.

Image	Model outcome	Visual interpretation	Classification	Limitation	Action
	No detection	A visible mesh of continuous parallel bars near the upper edge, with loss of concrete cover.	Suspected false negative	Poor lighting, strong perspective, partial obstruction; the rebar occupies a small portion of the frame.	Flag for engineer review; add low-light, wide-scene images to future training data.
	Detected, 0.72	Clear exposed reinforcement mesh; several continuous bars visible.	Likely true positive	One broad prediction covers a group of separate bars and some surrounding concrete.	Flag for assessment; do not use this output to calculate damaged area or repair quantities.
	Two detections, 0.44	Exposed steel visible in the central soffit area, but predicted regions are dominated by crack/spall geometry near the top.	Suspected false positive / poor localisation	The model may confuse cracks and damaged concrete with exposed rebar; low confidence supports uncertainty.	Human review; add crack- and spall-only negatives and more soffit-rebar examples.
	Four detections: 0.84, 0.66, 0.42, 0.30	Large damaged/spalled column area; no distinct continuous visible rebar is clear.	Suspected false positives	Vertical spalling, dark cavities and surface texture are confused with exposed reinforcement.	Human review; add damaged concrete without visible steel as negative examples.
	Detected, 0.44	Real exposed bars are visible lower in the soffit, while the mask mostly covers cracks and spalling above.	False positive plus missed localisation of the true bars	Confusion between crack/spall geometry and exposed steel; low confidence indicates uncertainty.	Human review; label soffit reinforcement correctly and add cracks/spalls as negatives.
	Two detections, 0.79 and 0.74	An exposed rebar grid is visible in the central spalled region.	Likely true positive	Large masks include substantial surrounding damaged concrete; the separate left-side 0.74 region shows no clear steel.	Flag the central area for assessment; treat the left-side region as a suspected false positive.
	Two detections, 0.82 and 0.41	Wide soffit view of a stair/landing area; the high-confidence mask spans most of the underside surface, while the smaller 0.41 region sits at the edge of the same element.	Likely true positive with heavy over-segmentation	The mask boundary follows the element, not the exposed steel, so it is not usable as damage extent.	Flag the soffit for close-range inspection; add close-range soffit images with tight masks to training.
	Two overlapping detections, 0.68 and one unscored region	A regular grid pattern of exposed bars is visible on the wall/column face where cover has been lost; the mask also covers adjacent spalled concrete and debris at the base.	Likely true positive	Mask over-extends into damaged concrete; duplicate overlapping boxes indicate weak non-maximum suppression.	Flag for assessment; do not derive exposed-steel area from the mask; tune NMS settings.
	One detection, 0.62	Distant view of a viaduct underside; a small patch of the soffit near the pier head is flagged. Visible steel cannot be confirmed at this image scale.	Unverifiable at this resolution	Visible steel cannot be confirmed at this image scale, so the flag can be neither accepted nor rejected.	Re-photograph the flagged area at close range; exclude distant wide-scene images or raise the confidence threshold.

6. Data and model improvement actions

- Add negative examples: cracks, rust staining and spalled concrete without visible steel -the dominant false-positive sources (images 3, 4, 5).
- Add close-range soffit and overhead reinforcement images with tight masks, to correct both the missed bars in image 5 and the over-segmentation in images 7 and 8.
- Add low-light, wide-scene and partially obstructed cases to recover misses of the type seen in image 1.
- Tune the confidence threshold and non-maximum suppression: overlapping duplicate regions (image 8) and low-confidence flags around 0.30–0.44 currently generate most of the review noise.
- Define an operational minimum image scale: distant views such as image 9 cannot be verified and should be re-photographed at close range.

7. Limitations and responsible use

The model cannot assess corrosion level, concrete strength, remaining bar diameter, structural capacity or repair urgency from an image alone. Performance depends on dataset coverage and is sensitive to lighting, shadows, occlusion, camera distance,

viewpoint and surface texture. Masks systematically include adjacent damaged concrete, so they must not be used to compute damaged area or repair quantities. Every prediction is an inspection flag for a qualified structural engineer, who remains responsible for any assessment, testing or intervention decision.

8. Conclusion

The prototype reaches 0.808 mask precision and 0.717 mask recall after 30 epochs and reliably locates clear cases of exposed reinforcement (images 2, 6, 8). Its weaknesses are systematic rather than random: confusion between cracking or spalling and exposed steel, coarse mask boundaries, and misses in low-light or wide-scene photographs. These are addressable through targeted data collection rather than a larger model, which makes the tool a credible first-pass screening aid for inspection photo sets — always under engineer review.